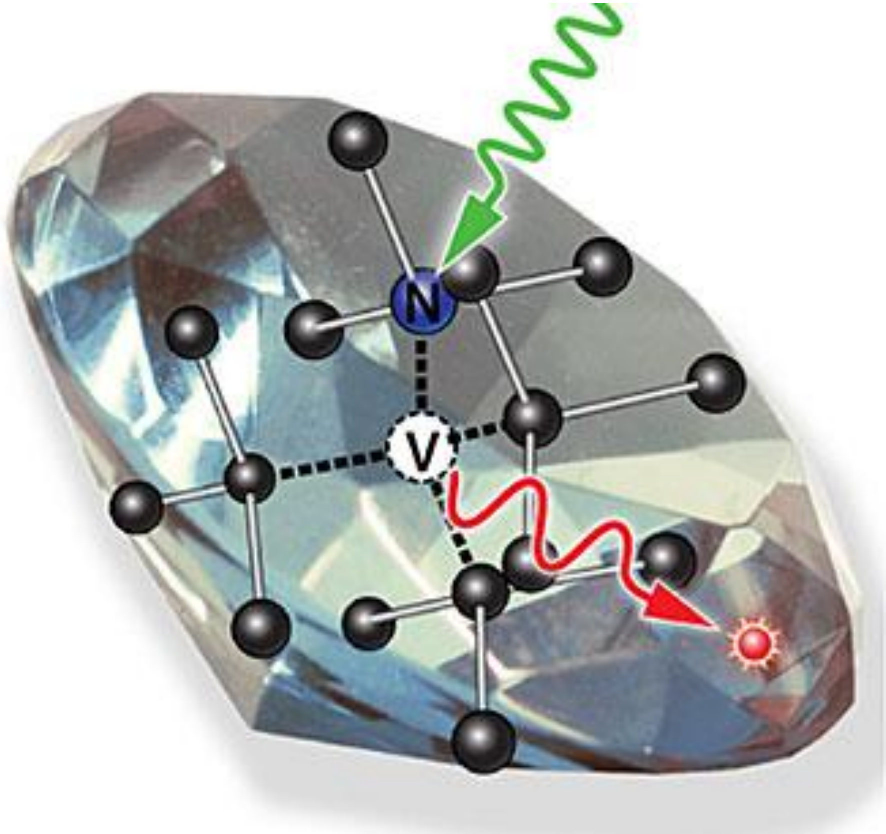
Qmet Tech

विज्ञान एवं प्रौद्योगिकी विभाग
DEPARTMENT OF
SCIENCE & TECHNOLOGY

National Quantum Mission

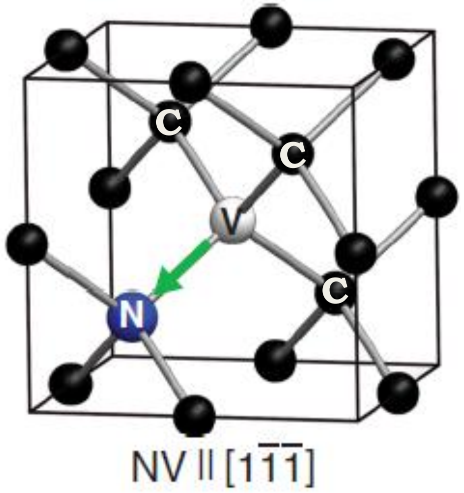
ECE IIITD

NV Centre Diamond Magnetic Sensors



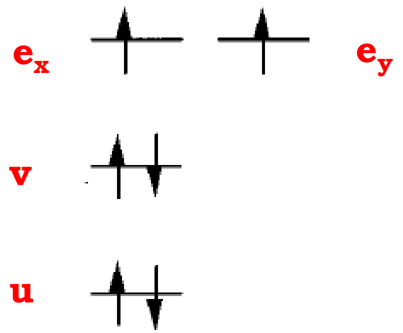
What Is NV Centre ?

- Substitutional nitrogen atom adjacent to a carbon vacancy in crystal diamond lattice.
- The 3 carbon atoms adjacent to the vacancy have only 3 bonds with other carbon atoms.
- So each of the 3 carbon atoms adjacent to the vacancy have 1 highly unstable **dangling electron**.
- The nitrogen atom has **1 lone pair of electrons**.
- NV^0 centre traps 1 extra e- from the nearby electron donor (another N impurity) to become NV^- centre with 6e-.
- The green arrow marks the NV symmetry axis.
- It acts as a **two hole system** where the 2 e- from the lone pair decide the spin triplet or singlet.



Diamond lattice containing an NV centre

[John F. Barry et al, Rev. Mod. Phys., 0034-6861/2020/92(1)/015004(68)]



Ground state spin configuration of 6e- NV centre

[A. Lenef et. al. Physical Review B, Volume 53(20), 0163-1829/96/53(20)/13441(15)]

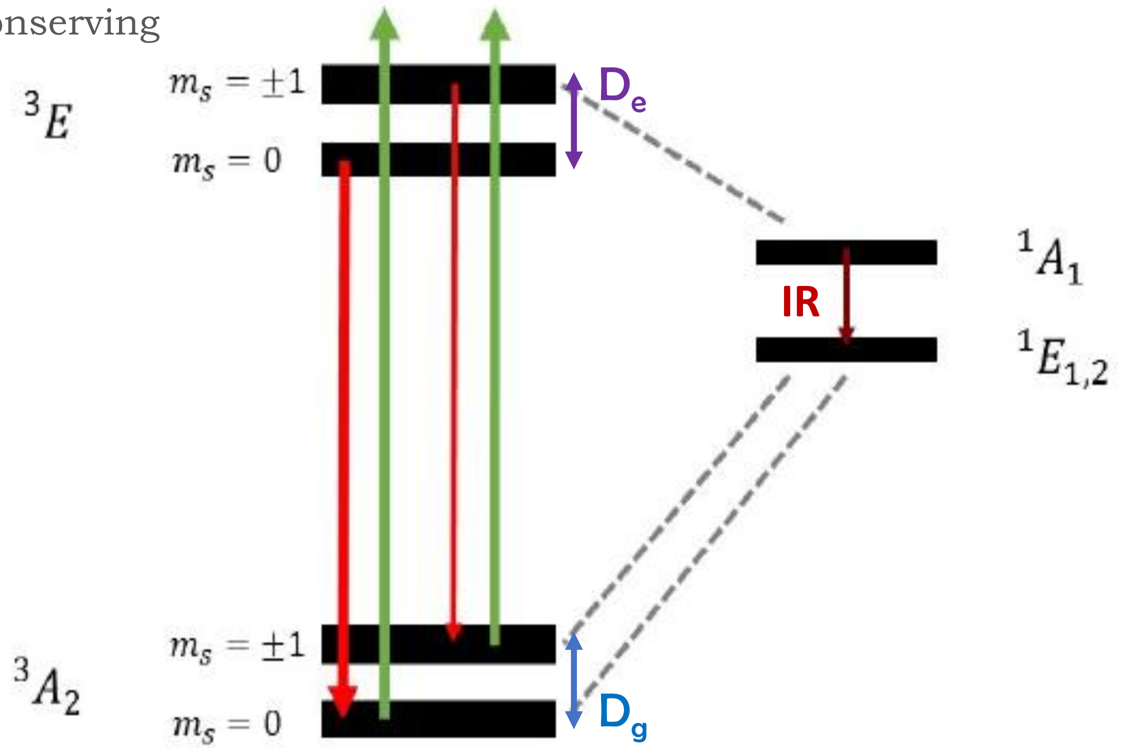
Nitrogen Vacancy Energy Level

Green Laser Pump

Excitation = always spin conserving
Decay = sometimes spin conserving

- Green light (532 nm) excites electrons from ground state triplet ^3A to excited state triplet ^3E
- e^- with $m_s = 0$ drop straight back down via radiative path emitting bright red fluorescence.
- Most of the e^- with $m_s = +/-1$ divert sideways non-radiatively via **intersystem crossing** into a singlet shelving state $^1A_1$
- Then from $^1A_1$ it decays to another singlet state $^1E_{1,2}$ emitting infra-red (1042 nm).
- And finally from $^1E_{1,2}$ it decays non-radiatively to $m_s = 0$ and $m_s = +/-1$ ground state level with equal probabilities.
- Continuous green light automatically **polarizes** almost all spins into the bright $m_s = 0$ starting state.

m_s = electron spin projection quantum number



Energy level diagram for NV− center in diamond

Ground state zero field splitting energy, D_g = 2.87 GHz

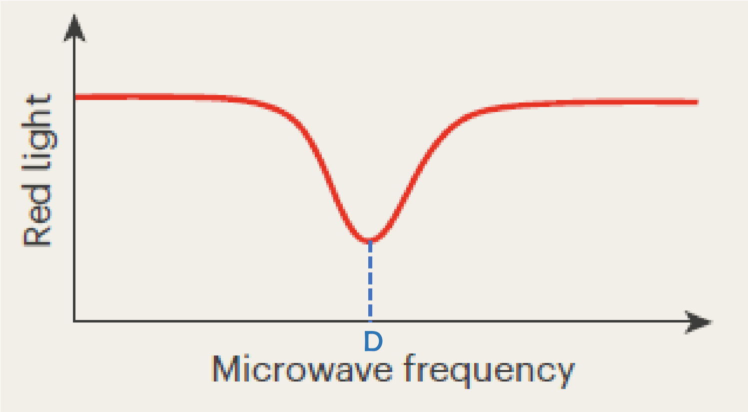
Excited state zero field splitting energy, D_e = 1.41 GHz

Rosenzweig, Yossi. (2016). The Physics of Nitrogen-Vacancy Color Centers in Diamonds and their Interaction with External Fields.

Nitrogen Vacancy Energy Level

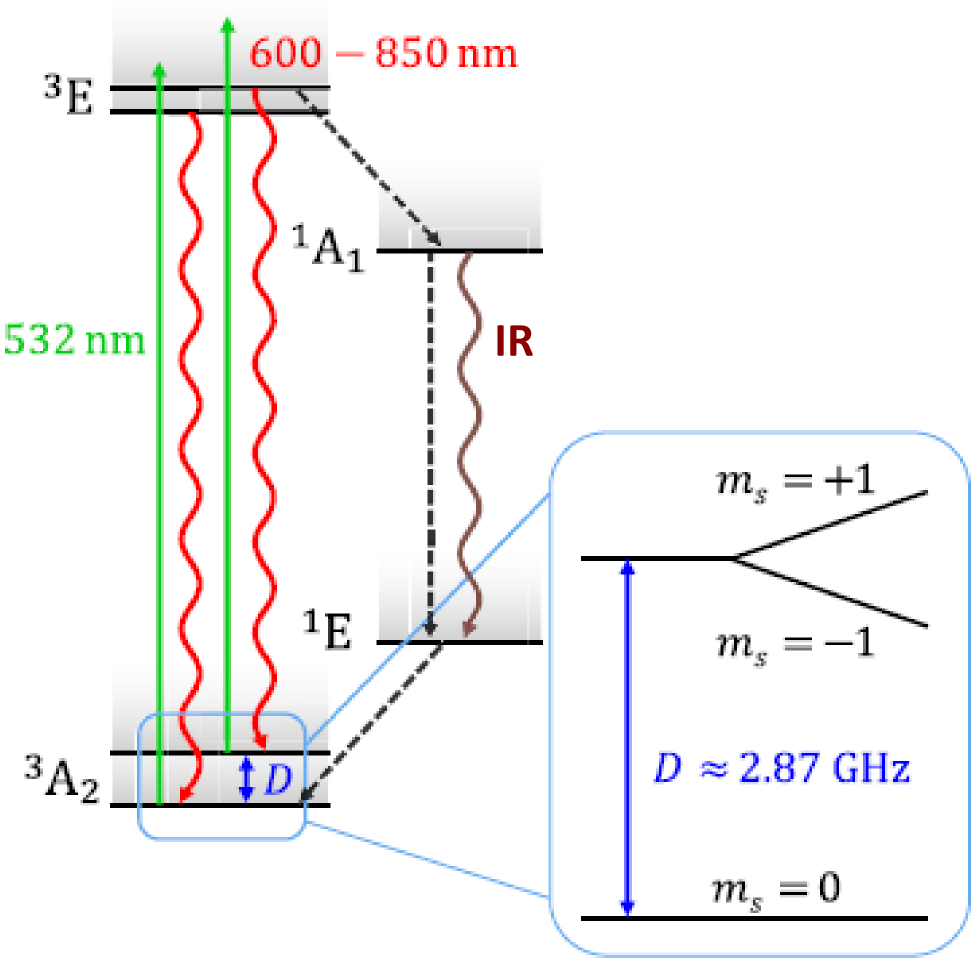
Microwave Signal

- An onboard MW antenna broadcasts and sweeps microwave frequency around 2.87 GHz.
- When the MW frequency matches the ground state energy gap, it flips spins from $m_s = 0$ to $m_s = +/-1$
- Diverted spins take the singlet shelved dark path causing a dip in the red light intensity.



Dip in the red fluorescence at spin flipping frequency

[N. Savage, Nature, Vol 591, doi.org/10.1038/d41586-021-00743-3]



Energy level diagram for NV– center in diamond

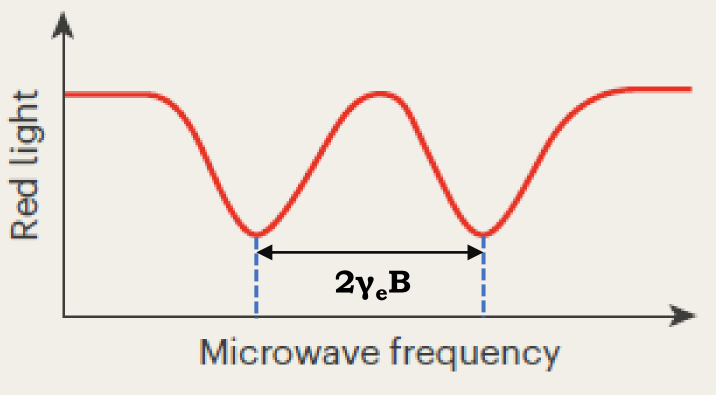
[John F. Barry et al. , Rev. Mod. Phys., Vol 92, 0034-6861/2020/92(1)/015004(68)]

Nitrogen Vacancy Energy Level

ODMR

γ_e = gyromagnetic ratio (28 Hz/nT)

- External low frequency magnetic field (B_z) is applied along the NV axis.
- **Zeeman shift:** MF induces an energy shift in the ground triplet state for m_s = +/-1
- Fluorescence dips occur at two distinct frequencies.
- The separation between these two dips directly related to the magnetic field strength.



Dip in the red fluorescence at spin flipping frequencies

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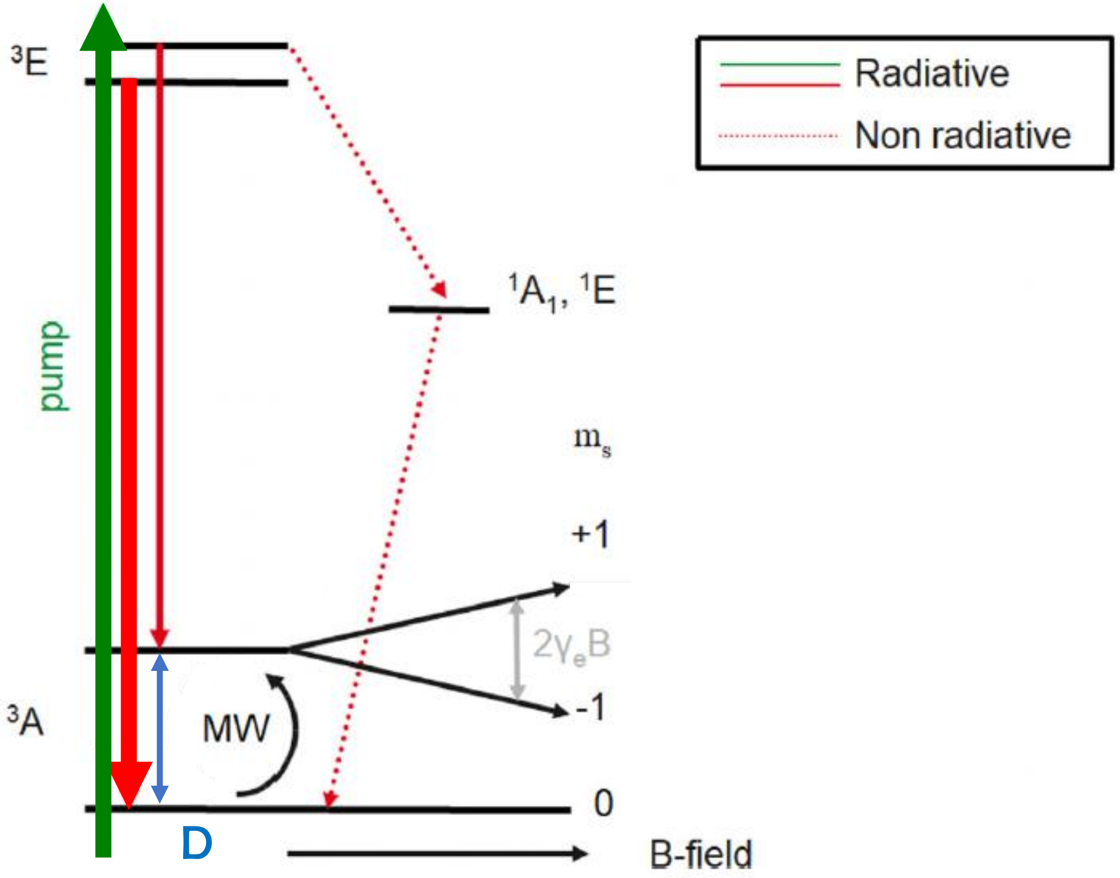


Energy level diagram for NV− center in diamond

[James L. Webb et. al., Appl. Phys. Lett. 114, 231103 (2019); doi: 10.1063/1.5095241]

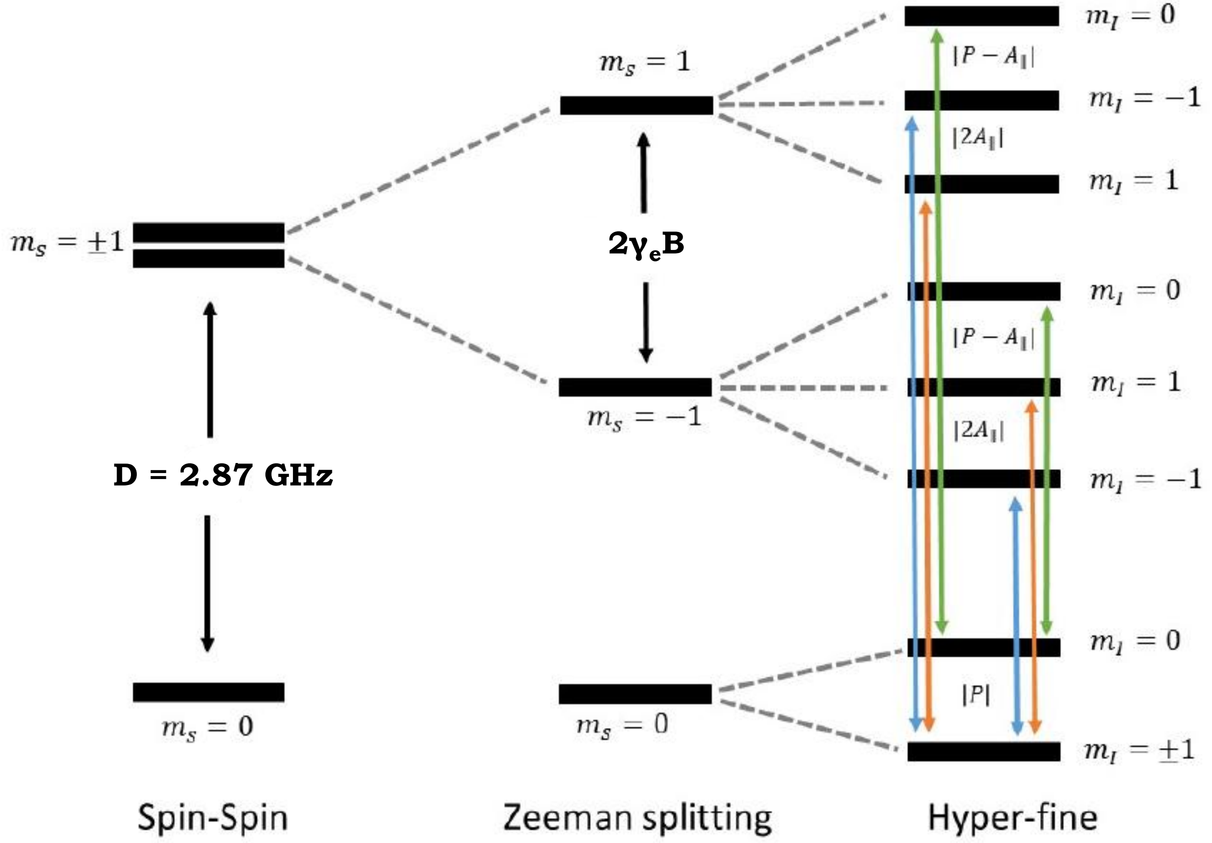
Zero field splitting energy (D = 2.87 GHz)

Nitrogen Vacancy Energy Level

^{14}N hyperfine structure

- Electron spin (S=1) couples with nuclear spin of adjacent N-14 atom (I=1)
- 3 fold splitting in each of the Zeeman levels forming a 9-state system.
- Each of the two main magnetic dips further split into three closely spaced mini dips.
- The green line shows transition from (m_s = 0; m_i = 0) to (m_s = +/-1; m_i = 0).
- The orange and blue lines show similar transition but with m_i = +1 and m_i = -1 respectively.

P = -4.95 MHz
A_{II} = -2.16 MHz



Ground state energy levels and allowed transitions in NV- centre

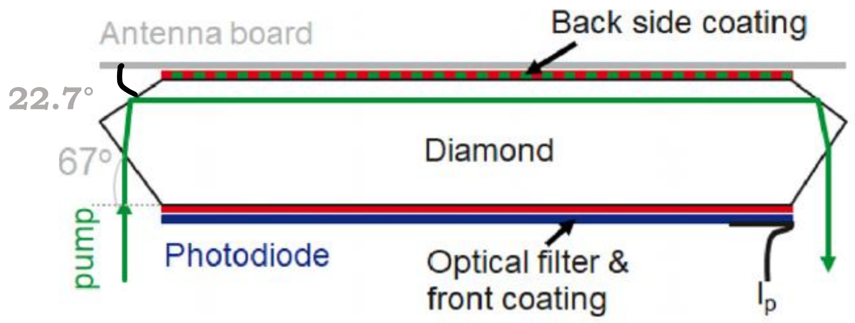
Rosenzweig, Yossi. (2016). The Physics of Nitrogen-Vacancy Color Centers in Diamonds and their Interaction with External Fields.

OPTICAL AND CRYSTAL DESIGN

Special Diamond Cuts

- **67° Cut (Brewster Angle):**
  - minimizes reflection loss.
  - ensures maximum transmission of green laser pump.

- **22.7° Cut:**
  - directs the laser beam laterally across the width of the crystal.
  - excites maximum number of NV centres.



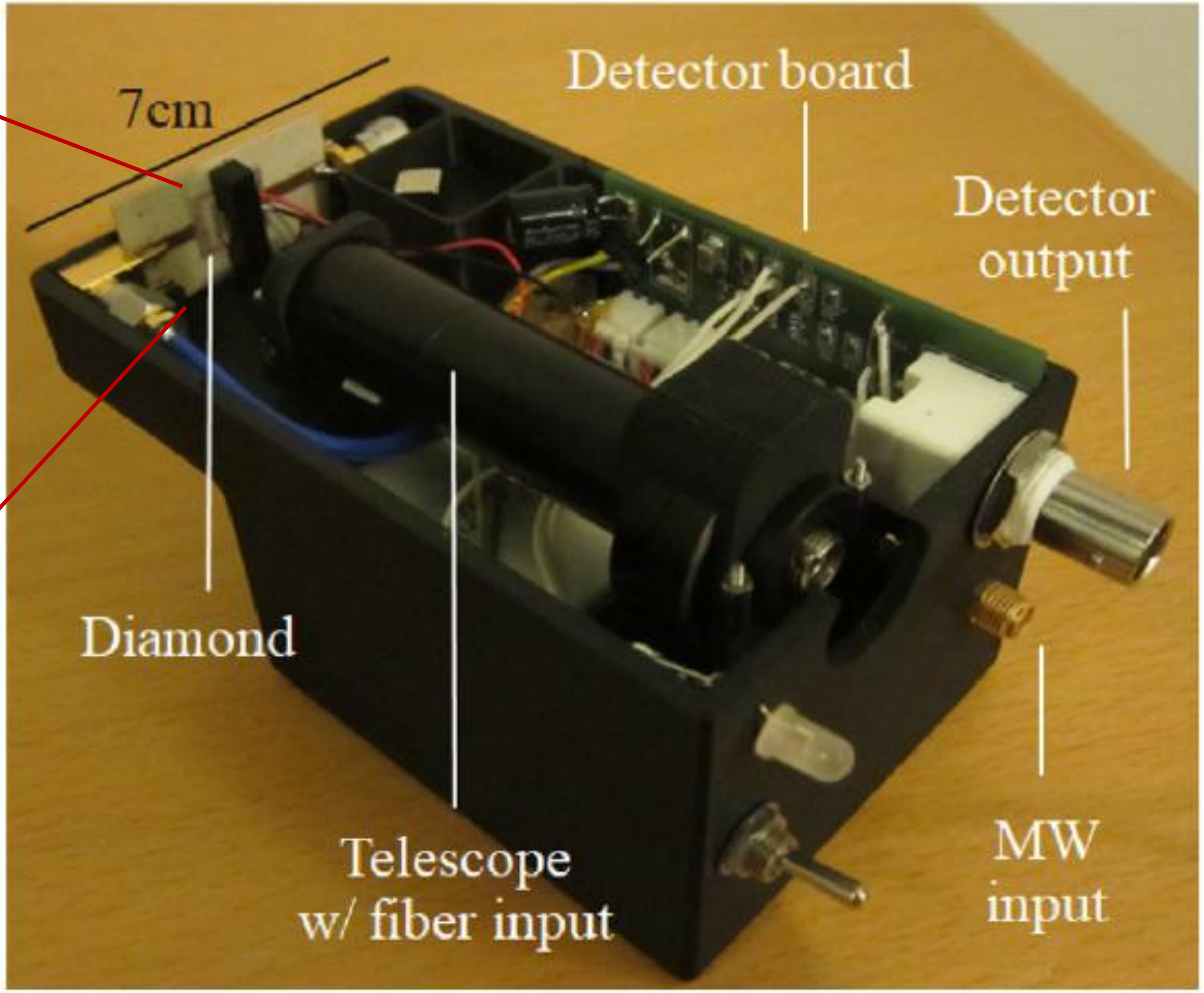
Schematic of the specially cut diamond crystal

Optical Coating and Filtering

- **Coatings:**
  - Back side facing the MW antenna is layered with highly reflective (R>99%) for wavelength 500-800 nm.
  - Front side has split coating. It is **highly reflective (R>99%)** for **532 nm** green light but **anti-reflective (R<1%)** for **600-800nm** red fluorescence.

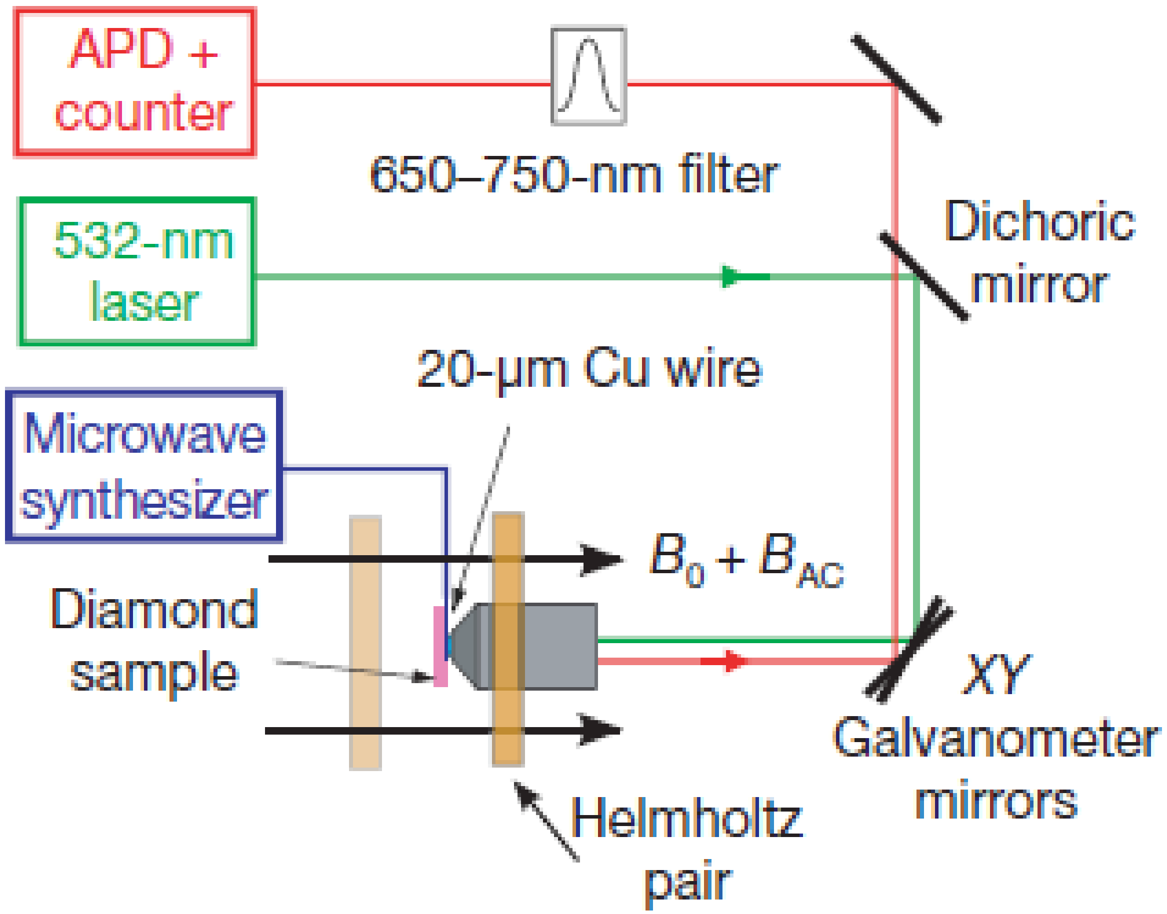
- **Filtering and Detection:** Optical long pass filter (550 nm) rejects stray green light so that photodiode can measure red fluorescence effectively.

Optical and crystal design



Handheld NV diamond magnetometer

Electronic Spin Echo Measurement



Confocal Microscopy setup

Electronic Spin Echo Measurement

AC Magnetic Field Sensing

- **Spin Echo**: Recovery of the signal with spin information.
- **Spin Echo Interval**:
  - Total free evolution time during which the NV spin accumulates phase from AC magnetic field.
  - Time difference between the two π/2 pulses.

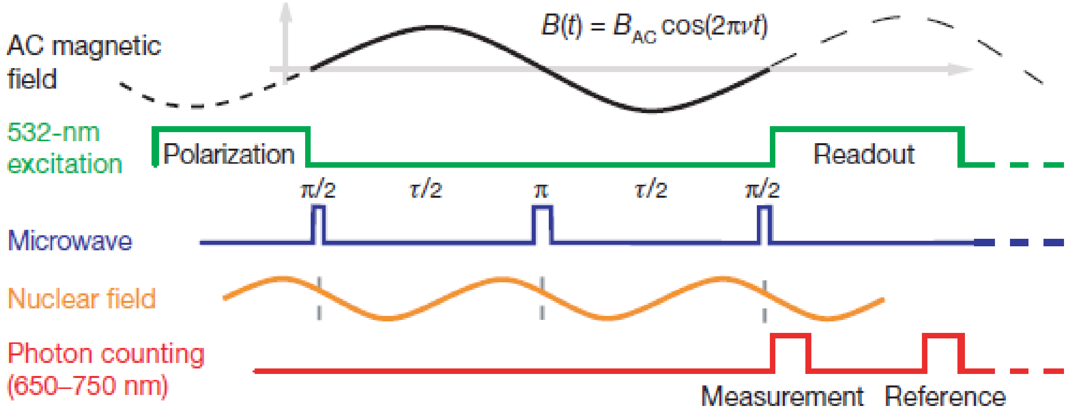
- **Polarization (1us):** A green laser pulse polarizes the spin into $m_s = 0$ ground state.

- **π/2 pulse:** MW pulse creates coherent superposition of $m_s = 0$ and $m_s = 1$ states. $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

- **Free Evolution (τ/2):** The NV spin accumulates phase under AC magnetic field and ^{13}C nuclear spins.

- **π refocusing pulse:** Swaps the spin ($|0\rangle$ to $|+1\rangle$) to cancel the static noise.

$\tau = 1/\nu$



Optical and microwave spin echo pulse sequence

J.R. Maze et. al., Nanoscale Magnetic Sensing in diamond, Nature-Vol 455 | 2 October 2008 | doi:10.1038/nature07279

Electronic Spin Echo Measurement

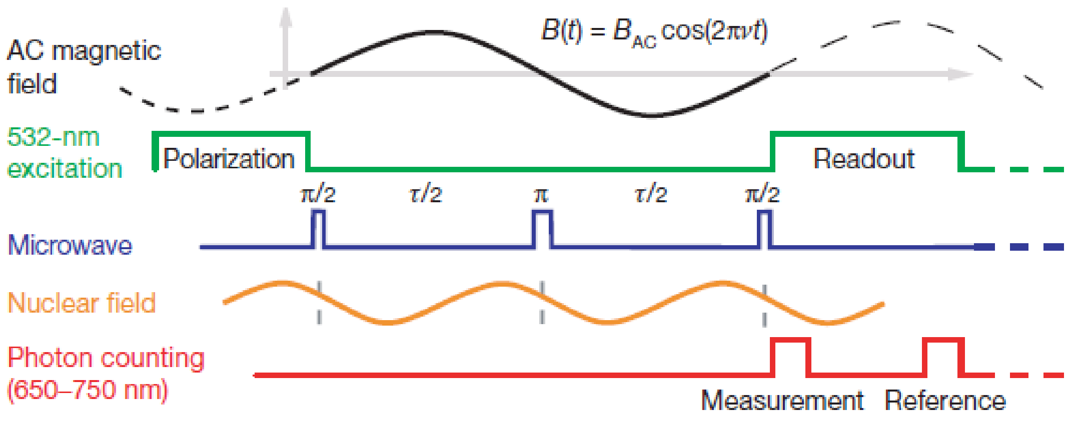
AC Magnetic Field Sensing

- **Free Evolution (τ/2):**
  - Phase accumulated here gets added to the phase from the first half as the field is oscillating.
  - But the phase due to static noise gets cancelled.

- **π/2 readout pulse:** The accumulated phase is converted into a population difference.

- **Readout (3us):**
  - The number of photons is counted during the measurement window.
  - The measured fluorescence is dependent to the probability of NV spin in m_s=0 state which is inturn dependent on the accumulated phase.

$\tau = 1/\nu$



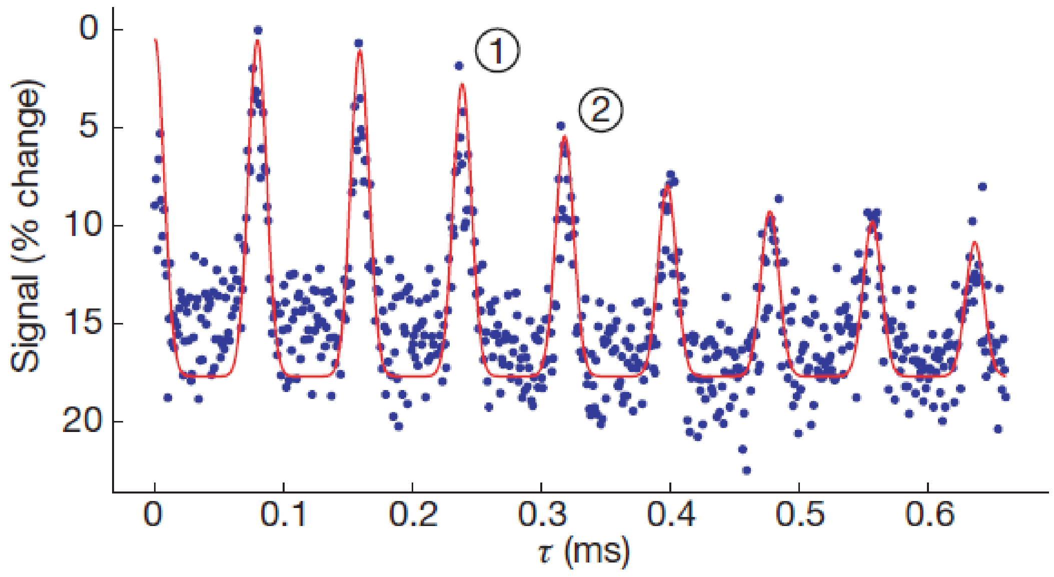
Optical and microwave spin echo pulse sequence

J.R. Maze et. al., Nanoscale Magnetic Sensing in diamond, Nature-Vol 455 | 2 October 2008 | doi:10.1038/nature07279

Electronic Spin Echo Measurement

Peaks and Valleys

- Normalized spin echo signal (corresponding to fractional change in NV centre fluorescence) over time.
- Oscillating peaks and valleys are caused by the NV spin interacting with surrounding bath of ^{13}C nuclear spins.
- **Peaks** = Revivals
  Valleys = Collapses
- Revivals occur periodically at half the rate of the **Larmor frequency** of the ^{13}C nuclei.
- Experiments are conducted at the revival peaks as the NV spin decouples from the magnetic noise of ^{13}C nuclei.



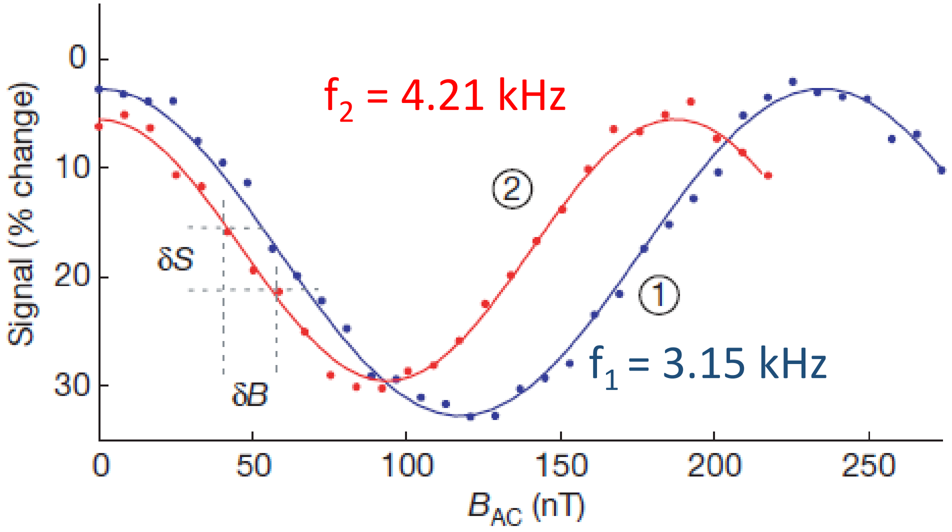
Electronic Spin Echo measurement over time

J.R. Maze et. al., Nanoscale Magnetic Sensing in diamond, Nature-Vol 455 | 2 October 2008 | doi:10.1038/nature07279

Electronic Spin Echo Measurement

Sensitivity of magnetometer

- Normalized spin echo signal as a function of amplitude of external AC magnetic field.
- With increasing field amplitude, NV spin accumulates phase due to time varying Zeeman shift.
- The phase accumulation is read out as a periodic change in the detected fluorescence.
- Magnetometer is most sensitive at the point of maximum slope (small change in magnetic field).
- **Sensitivity** is limited by uncertainty in measurement of spin echo signal (noise).
- Cosine behaviour of signal can be changes to a sine behaviour by adjusting the phase of microwave signal by 90°.



Electronic Spin Echo measurement with external AC magnetic field

Sensitivity: $$\delta B_{\min} \approx \frac{\sigma_S^N}{dS_B/dB_{AC}}$$

σ_S^N : noise

$\dfrac{dS_B}{dB_{AC}}$: slope

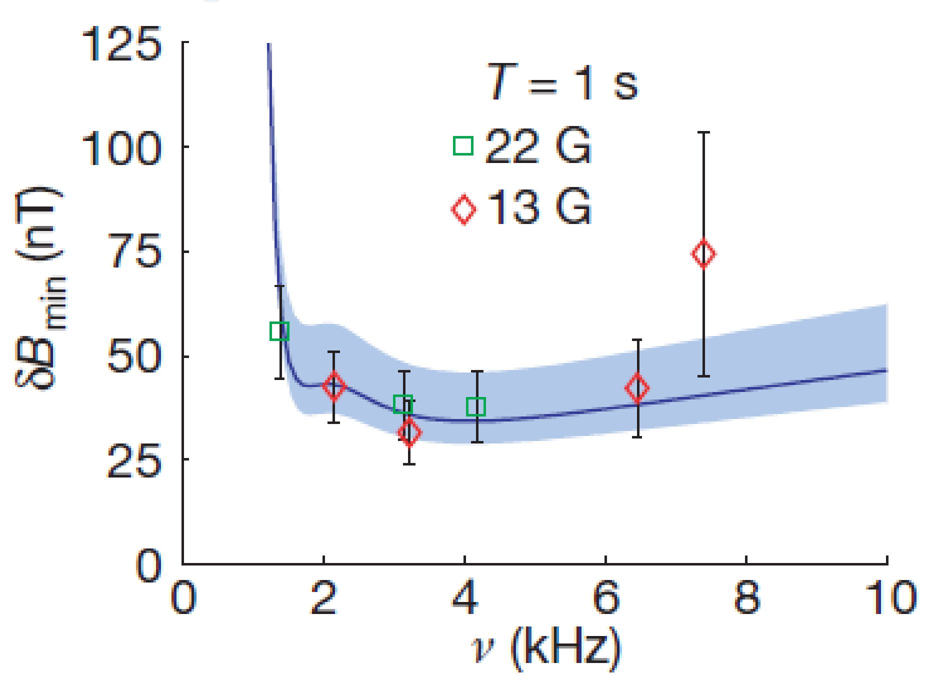
Sensitivity V/S AC Frequency

Results

- **Input Parameters**:
  - AC magnetic field frequency ranging from 1-10 kHz.
  - Measurements are taken at two different static magnetic fields **B_0 = 13 G** and **B_0 = 22 G**
  - Averaging time is T = 1s

- **Output**: Minimum detectable AC magnetic field

- **Observations**:
  - Sensitivity is lowest at around 3 kHz.
  - The minimum detectable field is about 30 nT.
  - At low frequencies, the sensitivity is very poor as the spin echo interval $\tau = 1/\nu$ becomes very long and NV spin decoheres before echo is formed.



Minimum detectable AC field over a range of frequencies

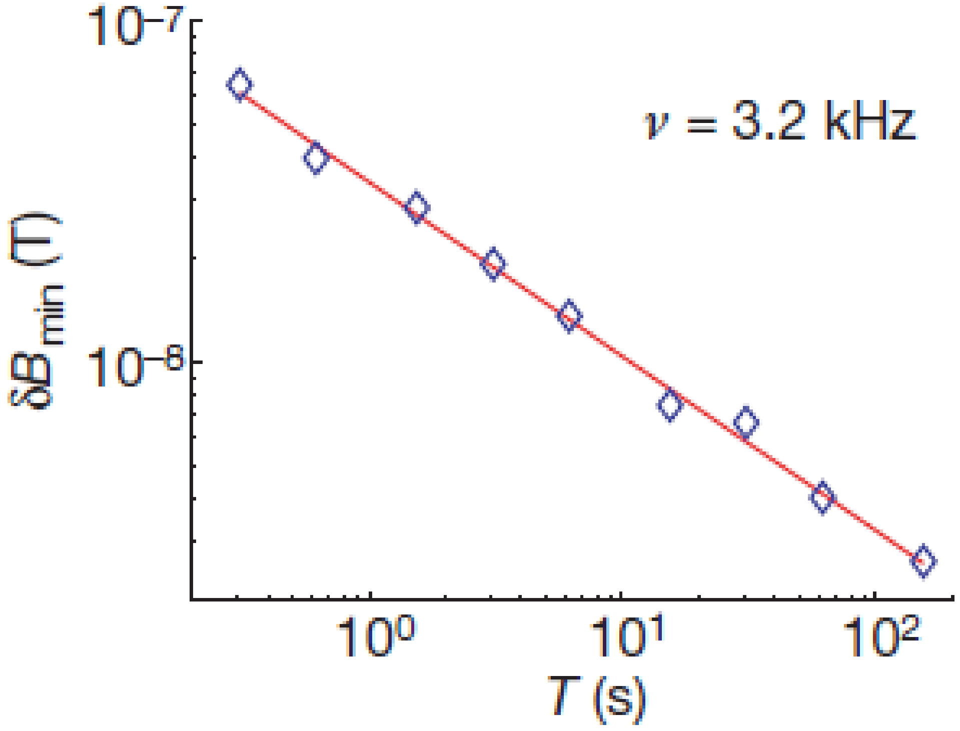
Sensitivity V/S Averaging Time

Results

- **Input Parameters**: AC magnetic field at ν = 3.2 kHz and B_0 = 13G.
- **Output**: Minimum detectable AC magnetic field on logarithmic scale.
- **Observations**:
  - Sensitivity improves with averaging time.
  - After 100s of averaging, magnetic fields as small as few nanoteslas are resolvable.

$$\delta B_{\min} \propto T^{-\alpha}$$

$$\alpha = 0.5 \pm 0.01$$



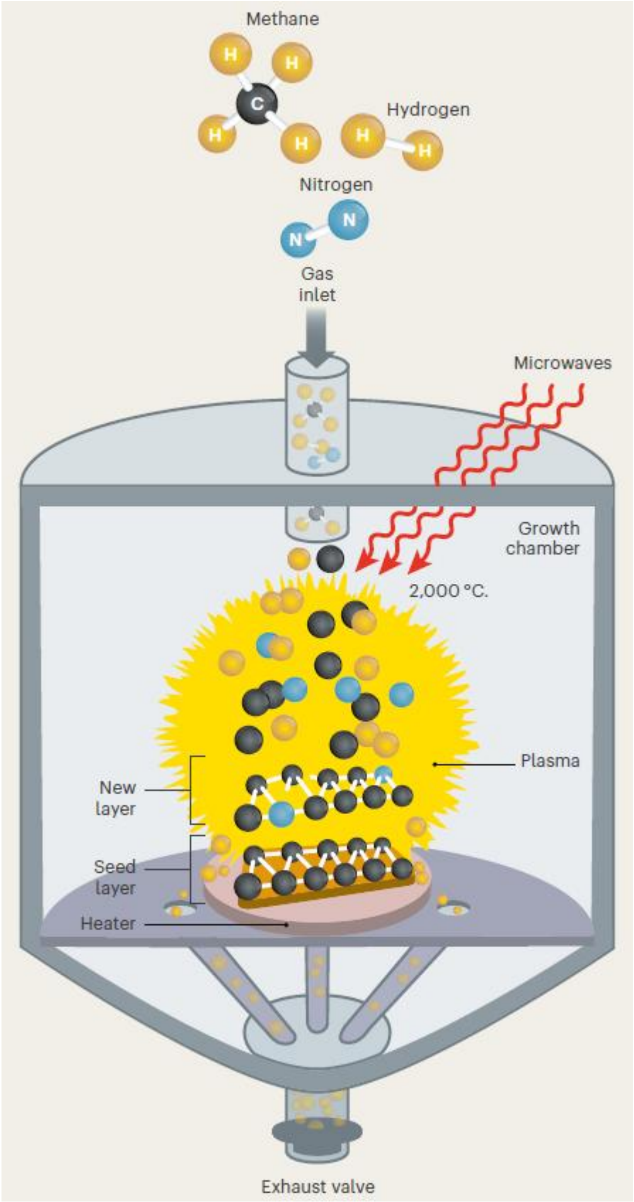
Minimum detectable AC field as a function of averaging time

N. Savage, Nature, Vol 591, doi.org/10.1038/d41586-021-00743-3

Growing Diamonds

1. Laying Down Layers

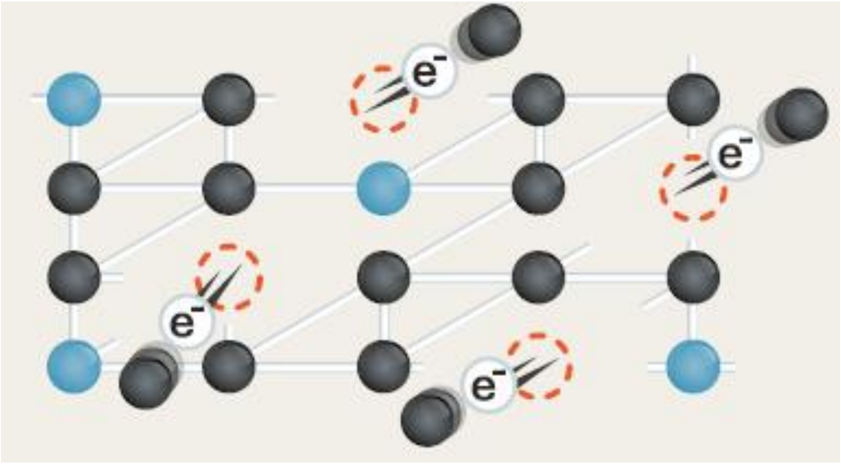
- A mixture of methane, hydrogen and nitrogen gas is superheated (2000°C) using MW to form plasma.
- Carbon atoms from methane as well as occasional nitrogen atom, settle out of plasma onto a heated seed layer.
- Hydrogen acts as a catalyst and modulates the process.



Growing Diamonds

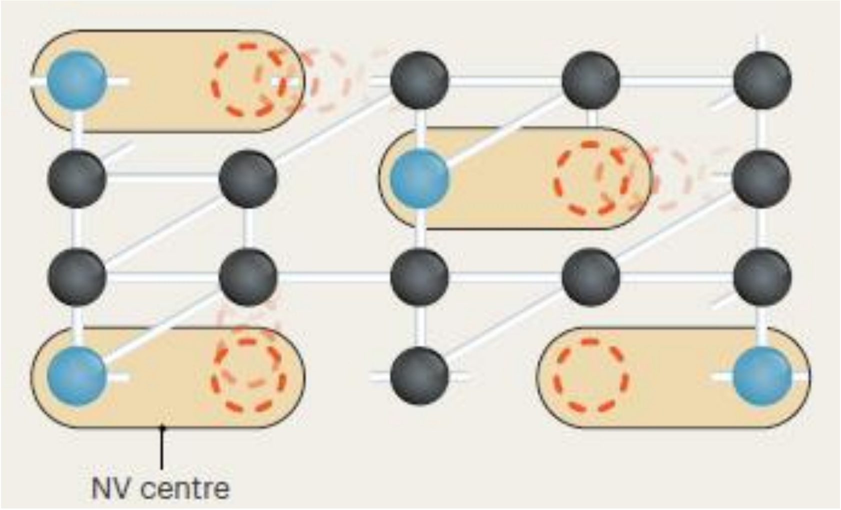
2. Bombardment

- The diamond film is hit with an high energy electron beam.
- It kicks some carbon atoms out of the crystal lattice to create empty spaces called vacancies.
- These vacancies are distributed randomly in the crystal lattice.



3. Annealing

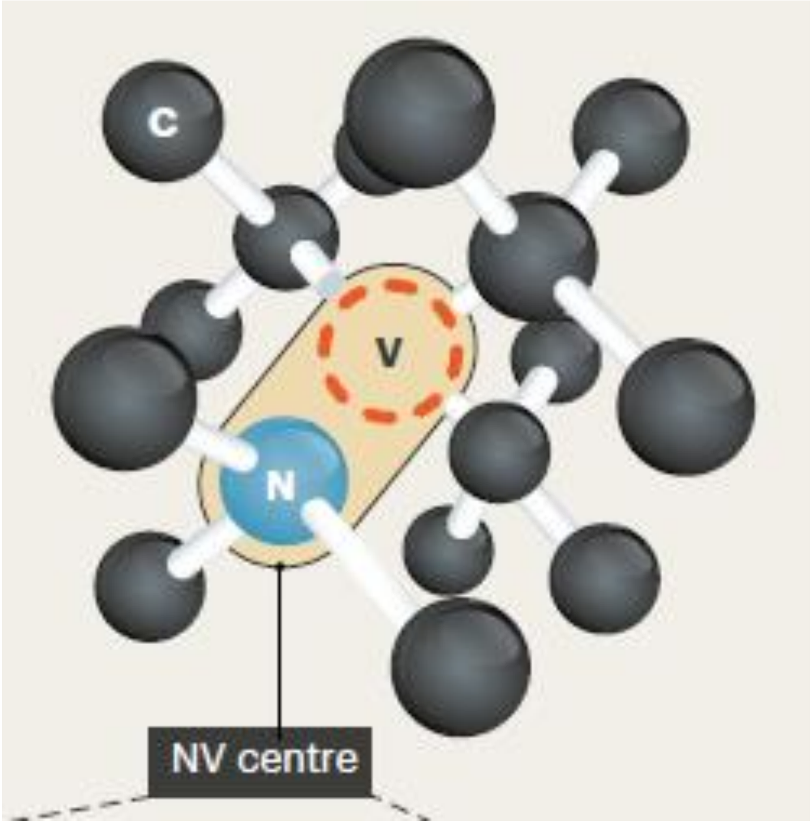
- Heating the diamond around 800° energizes the atoms causing the vacancies to move around.
- Many vacancies settle next to nitrogen atoms and form NV centres.



Growing Diamonds

4. Centres of Excellence

- The nitrogen atoms, vacancy and the neighbouring carbon atoms together form an “artificial atom”.
- This centre of excellence has its own quantum property known as spin.
- Spin can be pictured as a set of magnets with North and South poles.
- The arrangement of these magnets provides the spin state.



Experimental Setup

- **Diamond Sample**:
  - Company: Sumitomo
  - Fabrication technique: HPHT
  - Nitrogen conc. : 100 ppm or 1.76×10^{15} cm^{-3}
  - Size: 1.4 x 1.4 x 1 mm

- **Electron Irradiation**:
  - Device: High energy e- accelerator
  - Energy: 10 MeV
  - Diamond was water cooled in a copper tube to avoid vacancy-carbon recombination above 150°C.
  - After that diamond was sealed in quartz tube and annealed at 850°C.

- **Helmholtz Coils**:
  - A pair of Helmholtz coils was manufactured to provide static magnetic field to split the $|\pm1>$ degenerate levels.

Experimental Setup

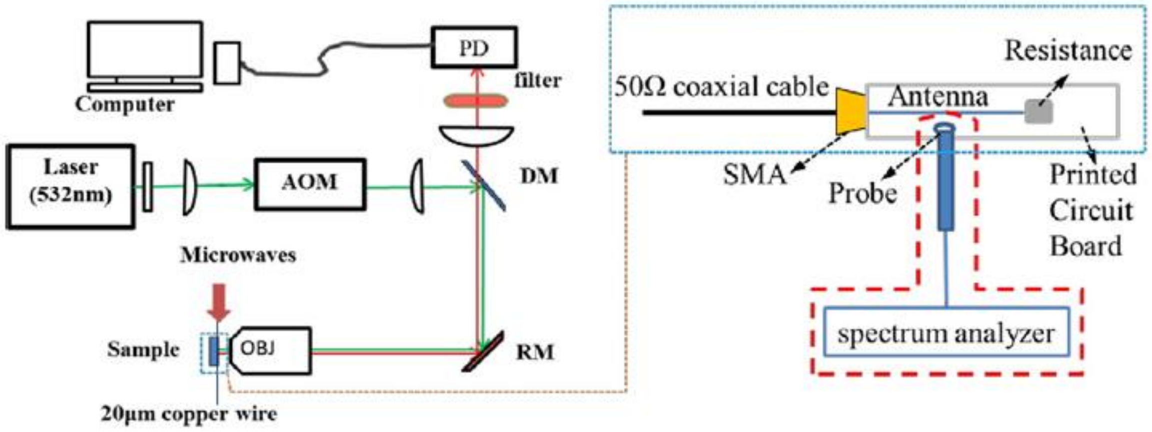
- **Laser (532 nm)**: Optical pump for green light.
- **AOM (Acousto-Optic Modulator)**: Gating the laser to create pulses.

- **DM (Dichoric Mirror):**
  - Reflects green light to sample.
  - Transmits red fluorescence to detector.

- **RM (Reflective Mirror)**: To steer the beam.

- **OBJ (Objective)**:
  - Focuses green laser onto the diamond sample.
  - Collects emitted red fluorescence.

- **PD (Photodetector)**: Detects red photoluminescence signal.
- **Antenna**: 20 um copper wire to deliver Microwaves.
- **PD (Photodetector)**: Detects red photoluminescence signal.
- **Long Pass Filter**: Allows only red fluorescence to pass through it.



Confocal ODMR setup

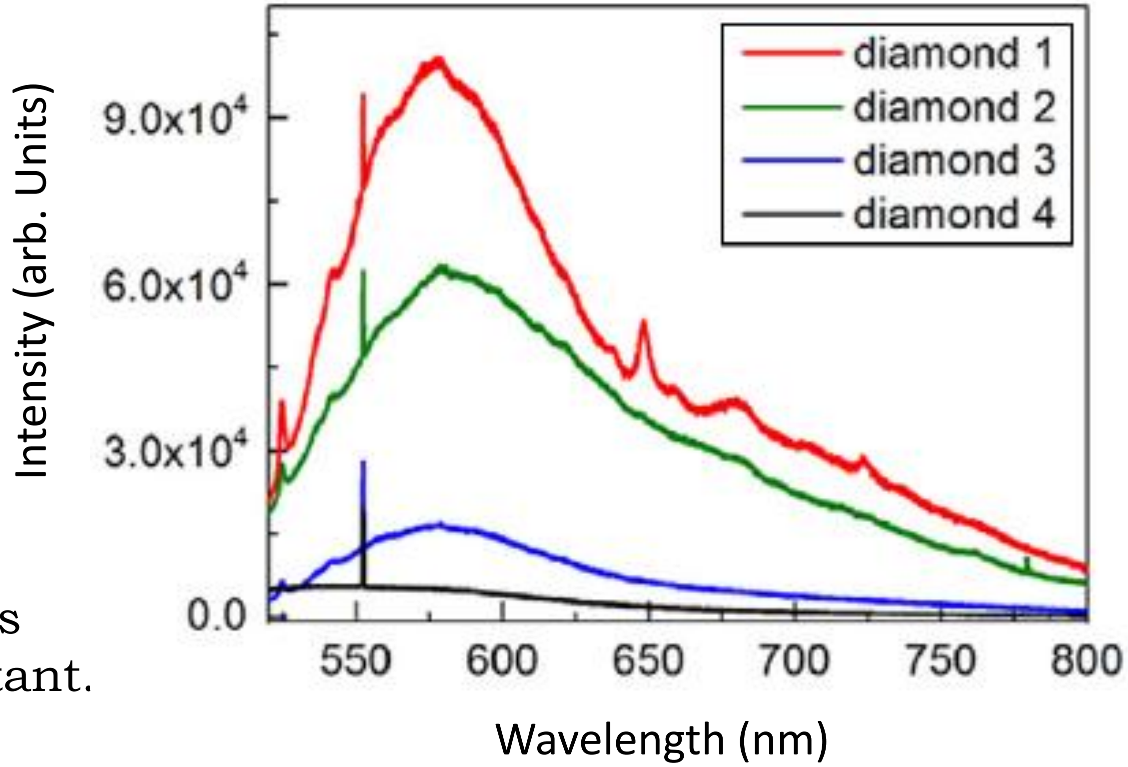
Spectral Analysis Of NV- Centre

Four different diamond samples were irradiated with electrons at an energy dose of 10 MeV for time :

Diamond 4 = 0 hour
Diamond 3 = 1 hour
Diamond 2 = 2 hour
Diamond 1 = 3 hour

Before Annealing

- A stable sharp peak at 552 nm across all the samples as the overall **carbon count** in the diamond remains constant.
- Some carbon atoms are physically knocked off from their normal positions due to electron irradiation.
- So another peak at 524 nm indicates these **carbon impurities** between the diamond lattices.



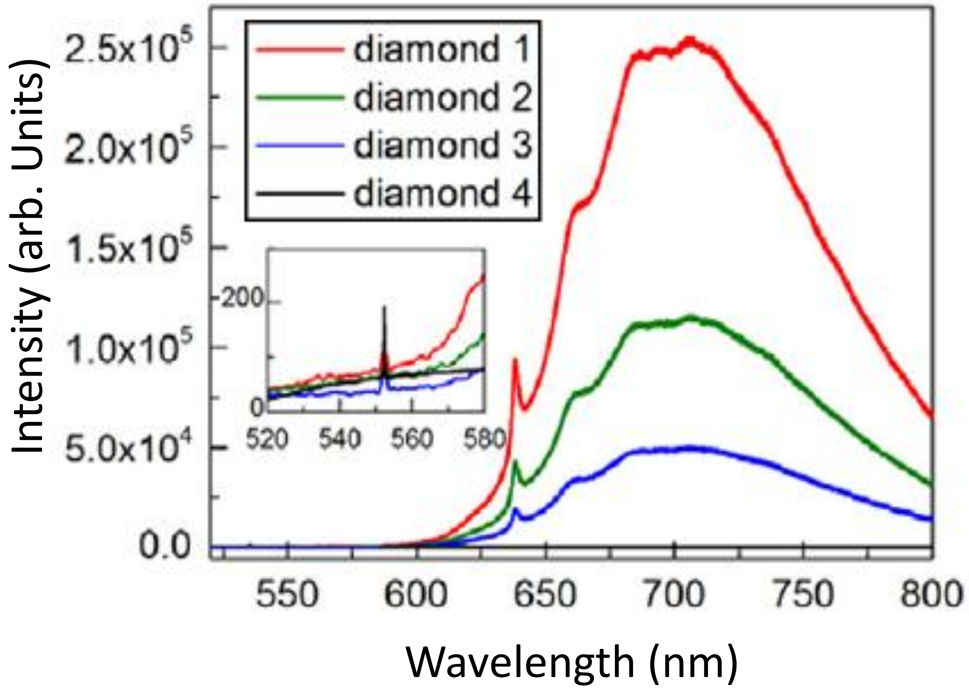
Room Temperature fluorescence spectrum of different diamond samples

Filiang Mu et al., Sensor Review (2017) Volume 37, Number 4 | doi:10.1108/SR-09-2016-0163

Spectral Analysis Of NV- Centre

After Annealing (850° C)

- The spectra is dominated by NV⁻ Zero Phonon Lines (ZPLs) located at 638 nm.
- **Zero Phonon Line:** Photon is emitted without losing any energy to surrounding diamond crystal lattice.
- The peak has a line width of roughly 1.5 nm **Full Width at Half Maximum (FWHM).**
- Peak increases progressively with longer electron irradiation time.
- Higher dose period of electrons successfully increases the conc. of NV- centres in the diamond.



Fluorescence spectrum of different diamond samples at 850° C

Diamond 4 = 0 hour
Diamond 3 = 1 hour
Diamond 2 = 2 hour
Diamond 1 = 3 hour

FWHM: Frequency or wavelength width where the intensity drops to 50%
Photon = Packet of light energy
Phonon = Packet of vibrational energy (heat or sound)

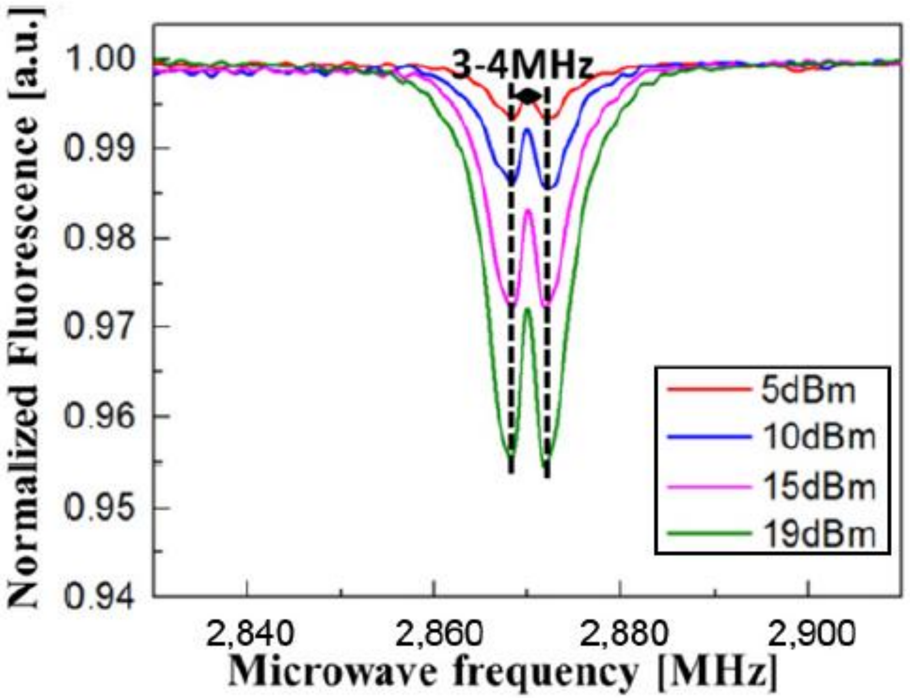
Electronic Spin Resonance

Results

- **Inputs**:
  - The MW power was set from 5 to 19 dBm.
  - MW frequency was scanned from 2.765 GHz to 2.975 GHz in steps of 1 MHz.

- **Output**: Normalized fluorescence.

- **Observations**:
  - At zero MW frequency no change in fluorescence is observed.
  - The fluorescence has 2 dips at 2.868 GHz and 2.872 GHz due to mechanical strain.
  - Increasing the MW power increases the depth of dips as it drives more NV spins.
  - Increasing the MW power also broadens the BW of dip as the spin transition saturates.
  - The 3.5 MHz splitting remains constant as it is due to strain and is independent of MW power.



Relative change in fluorescence with variation of MW frequency and power

Filiang Mu et al., Sensor Review (2017) Volume 37, Number 4 | doi:10.1108/SR-09-2016-0163

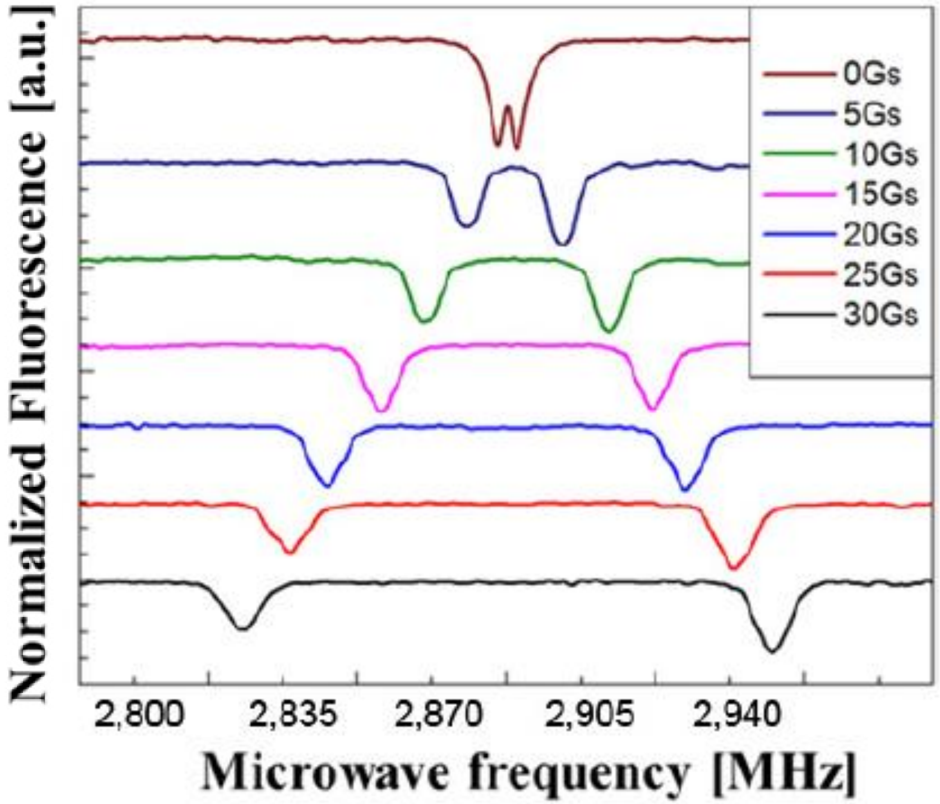
Electronic Spin Resonance

$$\nu_{\pm}(B_{NV}) = D \pm \sqrt{\left(\frac{g\mu_B}{h}B_{NV}\right)^2 + E^2}$$

Strain

Results

- **Inputs**:
  - The MW power was fixed at 19 dBm.
  - MW frequency was scanned from 2.8 GHz to 2.96 GHz.
  - Static MF was varied in steps of 5 G from 0 G to 30 G.
- **Output**: Normalized fluorescence.
- **Observations**:
  - Strain can be modelled as transverse EF.
  - As the MF increases, the two resonance dips split apart due to Zeeman effect.
  - The transition $\Delta m_s = +1$ shifts higher in frequency.
  - The transition $\Delta m_s = -1$ shifts lower in frequency.



Relative change in fluorescence with variation of MW frequency and static MF